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Aryabhatta Knowledge University, Patna

B.Tech. 6th Semester Examination, 2022

Time: 03 Hours

Code: 105601 ***Subject: Compiler Design*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer for any seven of the following : **[14]**

(a) Which of the following derivations does a top-down parser use while parsing an input string? The input is assumed to be scanned in left to right order.

(i) Leftmost derivation
(ii) Leftmost derivation traced out in reverse
(iii) Rightmost derivation
(iv) Rightmost derivation traced out in reverse

(b) What is the maximum number of reduce moves that can be taken by a bottom-up parser for a grammar with no epsilon- and unit-production (i.e., of type $A \rightarrow \varepsilon$ and $A \rightarrow a$) to parse a string with n tokens?

(i) $n/2$
(ii) $n-1$
(iii) $2n-1$
(iv) 2^n

(c) In a bottom-up evaluation of a syntax directed definition, inherited attributes can

(i) always be evaluated
(ii) be evaluated only if the definition is L-attributed
(iii) be evaluated only if the definition has synthesized attributes
(iv) never be evaluated

(d) Consider the grammar shown below : $S \rightarrow CC$, $C \rightarrow cC \mid d$. The grammar is

(i) LL(1)
(ii) SLR(1) but not LL(1)
(iii) LALR(1) but not SLR(1)
(iv) LR(1) but not LALR(1)

(e) The lexical analyzer takes _____ as input and produces a stream of _____ as output.

(i) source program, tokens
(ii) token, source program
(iii) Either of the two
(iv) None of the above

(f) In a compiler, when is the keywords of a language are recognized?

(i) During the lexical analysis of a program
(ii) During parsing of the program
(iii) During the code generation
(iv) During the data flow analysis

(g) How many tokens are there in the following C statement?
printf("j=%d, & j=%x", j, & j)

(i) 4
(ii) 5
(iii) 10
(iv) None of the above

(h) How is the parsing precedence relation defined?

(i) To delimit the handle
(ii) Only for a certain pair of terminal
(iii) None of the above
(iv) All of the above

(i) Given the following expression grammar : $E \rightarrow E*F \mid F+E \mid F$, $F \rightarrow F-F \mid id$. Which of the following is true?

(i) * has higher precedence than +
(ii) - has higher precedence than *
(iii) + and - have same precedence
(iv) + has higher precedence than *

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(j) Shift reduce parsers are

(i) top-down parser
(ii) bottom-up parser
(iii) may be top-down or bottom-up
(iv) None of the above

Q.2 Solve both questions : **[14]**

(a) Write a lexical analyzer program to identify strings, sequences, comments, reserved words and identifiers.

(b) Explain about the sources and criterions of code optimization as machine dependent and independent types.

Q.3 Solve both questions : **[14]**

(a) Construct the non-recursive predictive parse table for the given grammar and check the acceptance of input string "abfcg":

$S \rightarrow$ A
A $\rightarrow aB \mid Ad$
$B \rightarrow bBC \mid f$
C $\rightarrow cg$

(b) What is the relationship with lexical analyzer, regular expressions and transition diagram? Give an example.

Q.4 Solve both questions : **[14]**

(a) Explain the type system in type checker. Write the syntax directed definition for type checker.

(b) What is activation record? Explain its usage in stack allocation strategy. How is it different from heap allocation?

Q.5 Solve both questions : **[14]**

(a) Draw and explain the runtime memory organization static storage allocation strategy with pros and cons.

(b) Differentiate inherited and synthesized attributes with an example.

Q.6 Solve both questions : **[14]**

(a) Define dataflow analysis. List out the procedures to analyze the data flow of structured programs.

(b) Draw a block diagram of phases of a compiler and indicate the main functions of each phase.

Q.7 Solve both questions : **[14]**

(a) What is intermediate code? Translate the expression $(a+b)/(c+d)*(a+b/c)-d$ into quadruples, triples and indirect triples.

(b) Explain reducible and non-reducible flow graphs with an example each.

Q.8 Solve both questions : **[14]**

(a) What is machine-dependent optimization? Explain how peephole techniques function in this.

(b) Explain how type checking and error reporting are performed in compiler. Draw syntax tree and DAG for the statement: $a=(a*b+c)$wedge$(b+c)*b+c$

Q.9 Explain the following with suitable example : **[14]**

(a) Function preserving optimization techniques
(b) Reference counting garbage collectors
(c) Elimination of loop invariant variable
(d) Strength reduction

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